Vision Anomaly Detection Literature Review

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1 Introduction

Natural disasters represent some of the most destructive forces on Earth, posing significant threats to human life, infrastructure, and ecosystems globally. The devastating impact of such events has spurred growing interest in automated, intelligent monitoring systems capable of early detection and rapid situational assessment. The ability to identify precursory visual signals of disasters, such as ash plumes from volcanic eruptions, anomalous precipitation patterns, or hazardous particulate fog, is thus a critical research objective. One such field of computer science that encompass these requirements is visual anomaly detection. In order to develop a vision anomaly detection system suitable for the purposes of environmental monitoring, a comprehensive look into the current direction of the field with a focus on state of the art methods is needed. The field of visual anomaly detection encompasses anomalies whose modalities contain images and videos, where they have the potential reveal indications of hazards visible from miles away such as a volcanic eruption, an onset of heavy rainfall, or dense fog caused by particulate matter.

This synthesis was conducted for the purposes of understanding the fundamental structure involved in constructing visual anomaly detection methods and the challenges or assumptions involved such that it may serve as a launchpad into current and future works. Anomaly detection covers a plethora of subtopics, and a comprehensive look into the entire field is intractable. Thus, to serve as a launchpad for progressing the vision-related subsection of the field, this paper shall be a survey of recent techniques and publications that have had time to mature and evolve into seminal works. This review shall also be constrained to vision tasks where the model's inputs are primarily image or video feeds. First we will set the foundation by outlining the paper collection methodology categories into general domain applications and a deep dive into task taxonomy. Then followed by another taxonomy above the architecture of the models along with how a comparison and contrast of how each method justifies its broader category in the taxonomy. Each method is also associated with certain datasets and training paradigms which will be synthesized. Thereafter a discussion will be presented on observations and future direction. Lastly, all is wrapped up in the conclusion.

2 Survey Method

In an effort to discover the common tasks for visual anomaly detection, first a broad search for survey papers was conducted followed by a search for research papers. For each paper, a brief inspection of the abstract was applied to determine the primary scope of the paper. Each paper must have its primary focus be in regards to only vision anomaly detection methods, targeting image/video tasks. Duplicate papers were also removed. One retained survey mentioned medical imaging but lacked significant detail, so a dedicated medical survey [1] was added to cover that

domain. The contents of the remaining papers were reviewed and referenced papers were then investigated further. Seminal works were extracted from two or more overlapping references in the survey papers and are associated with high citation count, specifically, above 400 citations on Semantic Scholar.

GOOGLE SCHOLAR For GOOGLE SCHOLAR, the following search query was used ("visual" OR "vision") ("anomaly" OR "novelty") ("detection" OR "localization") ("survey" OR "review" OR "comparative analysis") and sort by relevance, the first 10 papers were considered.

LENS For LENS.ORG, the following search query was used (title:(Vision Anomaly Detection Survey) OR abstract:(Vision Anomaly Detection Survey) OR keyword:(Vision Anomaly Detection Survey) OR field_of_study:(Vision Anomaly Detection Survey)) and sorted by relevance (based on query matching score used in Elasticsearch), the first 10 papers were considered.

IEEE For IEEE, the following search query was used ("All Metadata":"visual" OR "All Metadata":"vision") AND ("All Metadata":"anomaly" OR "All Metadata":"novelty") AND ("All Metadata":"detection" OR "All Metadata":"localization") AND ("All Metadata":"survey" OR "Title":"review" OR "Title":"comparative analysis") and sorted by relevance, the first 10 papers were considered.

Use of AI AI tools were used in three capacities during the preparation of this review. First, as a semantic search aid to locate relevant passages within papers, with all retrieved content verified against the original source text. Second, as a sanity check for conceptual understanding of unfamiliar topics, with claims cross-referenced against the primary literature before inclusion. Third, for grammar, language and rephrasing of sections to improve comprehensibility. The models used were GPT-6 Astra and Claude Opus 4.5.

3 Domains

Each paper that is derived from the seminal works involves one or more domains of application. To be included under a given domain, a paper must either directly address it or demonstrate anticipated applicability through sufficient supporting evidence such as additional dataset evaluations, detailed observations, or a well-reasoned theoretical justification explaining how the method transfers to that domain. This bar exists because papers frequently reduce cross-domain applicability to a single speculative sentence. A passing mention of interest or motivation is not sufficient; concrete action or evidence is required. For example, a paper focused on defect detection in manufacturing clearly falls under Engineering and Technology, but a secondary claim of applicability to medical imaging is only counted if the authors substantiate it. The domains listed here do not encapsulate the entire field, but reflect those most prominently represented in the literature.

Industrial Covers the spectrum of industrial engineering and manufacturing, which is prominently featured in a majority of the seminal papers, along with robotics and autonomous systems.

Physical sciences Relates to the non-biological natural sciences such as material science, surface engineering, which in the context of anomaly detection, deals with surface defects and textiles. This section is also inclusive of astronomy, astrophysics, environmental science, and remote sensing.

Life sciences Encompasses the understanding of biological sciences such as biology and ecology.

Health sciences Differs from life sciences in that it deals with the applicability of knowledge to actualized care. Anomaly detection in this field deals with the diagnostics of medical data, medical imaging, and practical health care.

Social sciences Covers the scope of human behavioral sciences, economics, forensics, criminology, and public safety. Other interdisciplinary topics included under this field are transportation safety and traffic management.

4 Task Taxonomy

Visual anomaly detection serves many purposes ranging from satellite imagery, typically referred to as change detection, to industrial settings. The following is a list of common tasks visual anomaly detection has been used for. Solely based on survey paper count, public safety (people and roads) garnered the most attention with manufacturing coming second. Survey papers that maintain a primary focus will be considered for the task taxonomy. In the event that a paper's primary topic is too broad and coarse, it will not be included.

Table 1: Visual tasks derived from survey papers that have a domain specific focus.

Task	Domain	Description	Survey References
Industrial Manufacturing	Industrial	Quality assurance in manufacturing; detecting defects in production and manufacturing. Also used for visual inspection in routine maintenance for facilities. Automated Inspections.	[2][3][4]
Remote Sensing	Physical	Hyperspectral images, often comprised of hundreds of spectral bands, captured by satellite instruments to detect anomalies such as military operations, mineral deposits, maritime rescue, etc.	[5]
Urban Monitoring	Social	Urban environments involve detection of anomalies as it relates to public safety, society, roads, and traffic. Such anomalies include property damages, violence, robberies, traffic violations, etc. Typically involves people and crowds.	[6][7][8] [9][10]
Medical Imaging	Health	These tasks are related to medical imaging such as CT and MRI scans and detecting lesions.	[1]

Task	Domain	Description	Survey References
Inspection	Industrial	Inspections pertaining to infrastructure, specifically the survey paper focuses on contact lines.	[11]

4.1 Public Safety, People, Roads

Public safety anomalies deal with motor vehicles and city scapes. Anomalies may include traffic violations such as illegal U-turns, over-speeding, vehicles crossing stop lines or accidents [8]. Mayank Srivastava et al. referenced [7] Sultani et al. to describe that anomalies for this category also covers assault, arrest, arson, shooting, accident, burglary, explosion, shoplifting, fight, robbery, abuse, stealing, and vandalism. Santhosh et al's work describes the idea of three distinct anomaly types: Point Anomalies, Contextual Anomalies, and Collective Anomalies. **Point anomalies** are single isolated events like a car driving on the wrong side of the road. **Contextual anomalies** are anomalies conditioned on the context. To clarify the author's example, a biker traveling at a constant speed of 25 mph may be considered normal if the traffic surrounding the biker is also traveling around 25 mph. Conversely, if the traffic is at 5 mph and the biker travels at 25 mph, the biker's relative speed is significantly higher than the surrounding traffic and this may be considered an anomaly [8]. This is mirrored in Patrikar et al.'s and Zaheer et al.[6] work where they mention how 60 km/hr highway speeds are considered normal in most cases, but would be abnormal for a crowded street[12]. Lastly, **collective anomalies** are defined as an anomalies arising from a collective group of instances. To clarify, a stretch of road, where there is typically little traffic suddenly becomes inundated with many vehicles causing stop and go traffic. In this scenario, the anomaly is formed by the surge of individual vehicles piling up [8].

Detecting anomalies for public safety and surveillance is motivated by the impracticalities of manual monitoring on a massive scale [7][6]. However, through Srivastava et al.'s experiments for this task, robust detection still remains a challenge as the authors only report an accuracy of about 43% [7]. Duong et al. highlight some challenges as realtime constraints, time sensitive, high power consumption, feature extraction due to occlusion, overlap, clutter, noise, and classification ambiguity for intraclass and interclass similarity. Intraclass variation refers to differences within a class. For example, red and green apples are still apples. Conversely, Interclass is where some apples may appear identical to oranges in certain conditions. Essentially, the model gets confused where to draw the line for variations and objects bleeding into another class. [9].

Prior to 2019, most approaches addressed this problem by modeling a normal distribution from training data and flagging deviations in test samples as anomalous events [6]. Representative architectures explored in the literature include Convolutional Neural Networks (CNNs), Convolutional Neural Network-Long Short-Term Memory (CNN-LSTM) hybrid Three-Dimensional Convolutional Long Short-Term Memory (Conv3D-LSTM), autoencoder-Support Vector Machine (SVM) pipelines, ConvLSTM autoencoders, Convolutional Long Short-Term Memory (ConvLSTM) autoencoders, stacked Recurrent Neural Networks (RNNs), and Generative Adversarial Networks (GANs) [6]. As Patrikar et al. note, methods leveraging CNNs, LSTMs, GANs, and other deep learning architectures show strong promise, achieving superior accuracy across most surveillance scenarios [12].

4.2 Manufacturing

As noted by Cui et al., anomaly detection offers significant practical benefits in manufacturing and production by reducing reliance on human decision-making and specialized technical expertise, thereby lowering labor costs. By adopting AI-based systems, manufacturers can also improve consistency while reducing variance and subjective judgment between human operators. Safety considerations further drive the field of anomaly detection, particularly in maintaining strong focus on identifying material anomalies. Another factor that distinguishes manufacturing from many other domains is the unusual abundance of defect imagery [3].

Tao et al. and Li et al. describe two types of anomalies in this domain: textural and functional, both defined by their deviation from normal samples[2][3]. Textural anomalies contain low amounts of semantic information and typically manifest as small cracks, holes, or scratches. Functional anomalies carry a higher degree of semantic information, such as missing, misplaced, or misaligned components[2][4].

Cui et al. discussed some challenges being balancing the false positive alarm rate and missed alarm rate, where missed alarms lead to commercial loss from inferior products, but a high alarm rate will lead to increased costs in manual confirmations. Scarcity of labeled defect data and existing benchmarks are relatively simple and do not accurately reflect the complexity of actual production (also mentioned by Zhuo Li et al.[4]). Real world tasks may involve detecting impurities in edible oil, internal engine lining defects, wine impurities, bearing defects. Furthermore, difficulties may include small defect sizes, object appearance variability, and texture differences across materials[3]. Xian Tao et al. also mention another set of challenges which involves shifting the model's latent space to be compatible with human experience. For normal data, improper variations in illumination, perspective, scale, shadows, blur, etc. can result in significant differences that may cause false positives. Second involves a multiscale problem where a model needs to be able to handle defects spanning a few pixels to larger defects that cover the majority of the image. Lastly, scarcity of pixel-level labeled data makes precise segmentation of anomalous contours unlikely. Therefore the decision boundary will struggle to equal the ground truth distribution boundary[2].

Cui et al. surveyed how normalizing-flow-based methods are used to convert a simple distribution into a complex distribution by applying a series of invertible transformations such that distribution of normal samples is learned, conversely normalizing flows can also be used to learn the distribution of artificially augmented defects. Utilizing both is denoted as joint learning and improves the generalization ability of the model. Secondly, Representation-based methods apply a pre-trained model to extract image features for anomaly detection. These pre-trained, or foundation models garner massive amounts of data which allows them to capture a broad range of patterns and relationships. Enhanced by finetuning the model on specific tasks, may allow the model to quickly adapt to new domains and produce high-quality representations[3]. Lastly, Cui et al. mentioned multi-modal learning as a method to facilitate deeper understanding with the introduction of different modalities such as image and text[3]. Representative models spanning these paradigms include VAEs, GANs, PatchCore, PaDIM, SPADE, DRAEM, and CutPaste [3].

4.3 Hyperspectral

Hyperspectral images, are often comprised of hundreds of spectral bands, captured by remote sensing instruments. These spectral bands are similar to color channels in a traditional RGB image, but are not limited to visible light. Hyperspectral image detection is classified into spectral matching for known targets and anomaly detection for unknown targets. Anomalies are determined by low probability deviations from the hyperspectral background distribution. These deviations

may manifest as military targets, mineral exploration, maritime rescue, etc. Hao Fan et al. describe that hyperspectral data contains two types of information: spatial to determine position, and spectral which contains reflective characteristics [5].

The authors posed, in their survey of hyperspectral anomaly detection, three methods to address hyperspectral imaging anomalies: statistical methods, representation methods and deep learning methods. Statistical, defined by the authors, follows a specific Gaussian probability distribution where anomalies deviate from it. The logic is simple, but often fails to properly map complex background distributions. Representation based methods, utilize approximation by linear combination from known background elements in the dictionary. If the pixel can be approximated well it is considered normal, otherwise it is anomalous. Representation is more adaptable, but requires solving complex optimization problems, resulting in high computational complexity and insufficient real-time performance, with additional cost of manual parameter tuning. Lastly, deep learning methods, being the most recent, utilizes feature learning and learn the background from the data. Deep learning methods are capable in learning complex patterns but require a significant computation and data resources [5].

4.4 Infrastructure Inspection - Contact Lines

Overhead contact lines (OCLs) are electrified wires suspended above railways or roadways to supply vehicles with electric power. Power is transmitted through the contact between the vehicle's deployed pantograph and the contact wire. Because OCLs are continuously exposed to dynamic pantograph excitations, extreme weather, and environmental stressors, they are prone to localized anomalies that pose latent risks of failure if left unaddressed, even if there is no immediate power transmission disruptions [11]. In this context, an anomaly refers to either a deviation of a measured parameter-geometrical (e.g., contact wire height, stagger, residual wear), electrical (e.g., arcing between the wire and pantograph strip, hot spots), or mechanical (e.g., contact force, pantograph head acceleration) from its nominal range, or a defective condition of an OCL component such as cracks, looseness, abrasions, missing parts, deformations, or shifts in cross-span and contact line equipment [11].

Yu et al. describe the unique challenges of this class of anomaly detection task. Firstly, cameras mounted on high-speed trains are affected by sway and vibrations causing geometric measurement errors, blurry imagery, and compensation algorithms only factor in tilt but not pitch and yaw. Secondly, from the author's surveyed papers, imagery is in mostly similar positions and lighting which may not reflect real world environment variations. Real world cases suffer from the influences of weather, illumination, and complex backgrounds. Thirdly is the nature of anomaly detection, where some defects are extremely rare such as a crack on a swivel only occurring once on the Guangzhou railway during the last three years. Fourthly is the real time computation constraint, where the model is expected to output a response within a set time frame due to the travel speed of trains ranging from 120km/h to 350km/h. Fifth, perspective shifts can cause false positives, where from certain angles, any slight disengagement of contact points may visually appear still in contact. Lastly is the lack of datasets present for OCL tasks [11].

Yu et al. categorize computer vision approaches for OCL anomaly detection into stereo vision-based methods, which reconstruct 3D geometry via triangulation to measure parameter deviations, and object vision-based methods, which apply cascaded CNN pipelines, typically Faster R-CNN, for component detection and defect identification. To mitigate the scarcity of labeled defects, they advocate semi-supervised and unsupervised strategies such as autoencoders and GAN-based augmentation [11].

4.5 Medical

According to Cai et al., there are two types of anomalies as it relates to the medical field. Local anomalies and global semantic anomalies. Local anomalies are usually small lesions within healthy tissue. Global semantic anomalies are abnormalities that cover the entire image. [1].

Cai et al. identify several open problems in medical anomaly detection. Many widely used benchmarks such as Hyper-Kvasir and OCT2017 are "remarkably easy for anomaly detection", undermining their value for evaluation. The core reconstruction assumption is also fragile: a model trained on normal images may either generalize too aggressively and faithfully reconstruct anomalies (false negatives), or accumulate intrinsic reconstruction error on normal regions that masks true lesions (false positives). Latent space configuration remains dataset-dependent, with no principled way to select an optimal compression rate without test-time tuning, and the choice of distance function for measuring reconstruction error is similarly inconsistent across modalities, with no single metric dominating. Frozen ImageNet-pre-trained encoders prove surprisingly effective, often outperforming bespoke self-supervised methods but the path to improving on this baseline is unclear. Denoising diffusion probabilistic models (DDPMs) are constrained by their noise schedule, working only when the injected noise is sufficient to corrupt the anomalous region prior to reconstruction. Notably, simple intensity thresholding achieves the second-best segmentation performance on BraTS2021, outperforming more sophisticated DDPM-based pipelines. Finally, volumetric scans are typically reduced to 2D axial slices, discarding information along the third dimension [1].

Cai et al. benchmark thirty representative methods across seven medical datasets spanning five imaging modalities, finding that no single approach dominates across all settings. In the absence of pretraining, methods that learn to regenerate normal data prove more robust than those trained to discriminate synthetic pseudo-anomalies, and the simple autoencoder still remained a strong baseline. When synthetic anomalies are used, realism of the synthesis and a two stage training procedure (representation learning followed by a one-class classifier) outperform direct one stage discrimination. Most strikingly, by leveraging ImageNet-pre-trained weights, whether for distance measurement, input transformation, or direct feature extraction had yielded the strongest overall results, suggesting that the discriminative power of pre-trained features, rather than any particular detection paradigm, is the dominant factor in current medical anomaly detection performance [1].

5 Methodology and Techniques

Anomaly detection machine learning pipelines share commonalities which can allow it to be decomposed into modular building blocks which are referred to as techniques in this review. This perspective captures the larger picture and provides insights into new possible combinations of techniques or improving upon certain techniques. For instance, a hypothetical pipeline using a reconstruction technique can benefit from better reconstructive capabilities with the introduction of a pyramid technique which allows sampling from features at various layers in the network. Alternatively, improving on a technique would involve strategies such as optimizing for the spatial complexity. Optimization may include compressing the normal samples in a memory bank. In an effort to derive insights from the methodologies described in the papers, a taxonomy (Figure 1) was developed that grouped papers based on the principles used in their pipelines. A pipeline may be constructed from multiple techniques, so the techniques are not mutually exclusive. As an example, a pipeline can contain a memory bank method which contains a distribution of normal examples which is then used in a reconstruction method.

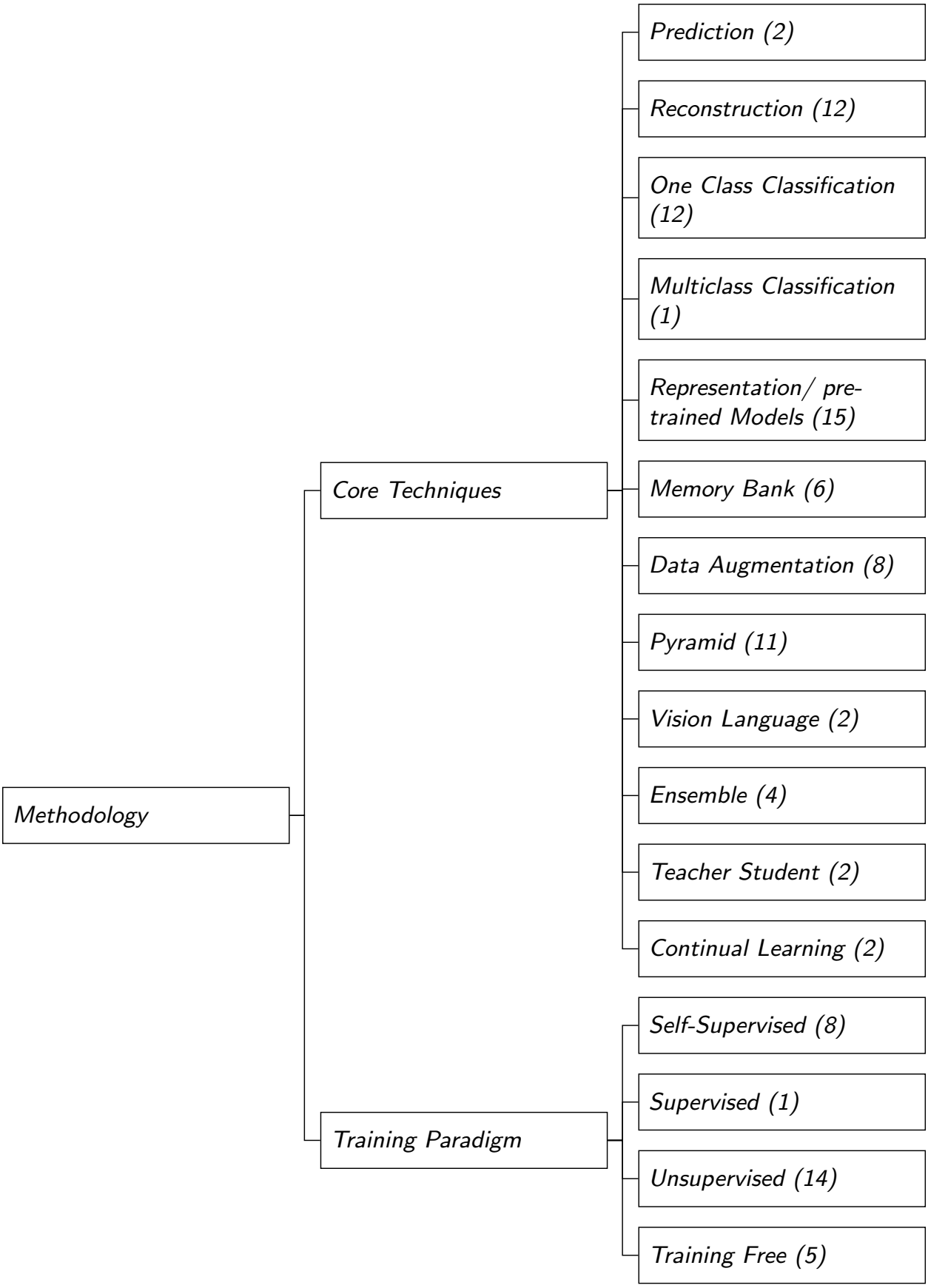


Figure 1: Taxonomy of the reviewed anomaly detection methods organized by training paradigm and core technique. Numbers in parentheses indicate how many surveyed papers fall into each category. A paper may use several core techniques, so counts are not mutually exclusive.

Table 2: Taxonomy of the reviewed anomaly detection methods organized by paper. Each paper (method) highlights the core techniques utilized and the training type along with which applicable domain it targeted.
[1] Papers that are not part of the seminal works

	Techniques												Training				Domain					
Method	Representation	One Class	Reconstruction	Pyramid	Data Augmentation	Memory Bank	Ensemble	Continual Learning	Teacher Student	Vision Language	Prediction	Multiclass	Unsupervised	Self-Supervised	Training Free	Supervised	Social	Engineering	Industrial	Health	Physical	Life
Future Frame Prediction [13]	✓										✓			✓			✓					
STAE [14]			✓								✓		✓				✓					
DRÆM [15]			✓		✓									✓				✓			✓	
Reverse Distillation [16]	✓	✓	✓	✓					✓				✓					✓				
Stacked RNN [17]	✓		✓	✓									✓				✓					
MemAE [18]			✓			✓							✓				✓	✓				
GANomaly [19]		✓	✓										✓				✓					
AE (SSIM) [20]			✓										✓						✓		✓	
AnoGAN [21]		✓	✓										✓							✓		
Spatiotemporal Autoencoder [22]			✓										✓				✓					
2D CNN AE [23]			✓										✓				✓					
RIAD [24]			✓		✓		✓							✓			✓		✓			
MaskVIT[1] [25]	✓		✓	✓									✓						✓			
PatchCore [26]	✓	✓		✓		✓									✓		✓	✓				
SPADE [27]	✓	✓		✓		✓									✓		✓	✓				✓
PaDiM [28]	✓	✓		✓											✓		✓	✓				
CutPaste [29]	✓	✓			✓		✓							✓				✓				
FastFlow [30]	✓	✓		✓									✓					✓				
CD-SVDD [31]		✓											✓				✓		✓	✓	✓	✓
GEOM [32]		✓			✓							✓		✓			✓					
PANDA [33]	✓	✓						✓					✓						✓	✓	✓	✓
CSI [34]		✓			✓		✓							✓			✓		✓	✓		
AnomalyGPT[1] [35]	✓			✓	✓	✓				✓				✓					✓			
IAD-GPT[1] [36]	✓			✓	✓	✓				✓				✓					✓			
AnomalyDino [37]	✓				✓	✓									✓			✓				
Uninformed Students [38]	✓			✓			✓		✓				✓					✓				
MIL [39]	✓															✓	✓					
PatchCoreCL[1] [40]								✓							✓				✓	✓		

5.1 Core Techniques

The following subsections presents a core strategy for anomaly detection. These strategies are best understood as conceptual building blocks that can be composed into pipelines or hybrid systems rather than as mutually exclusive alternatives. For instance, a reconstruction-based approach can be combined with a pyramid method, with each component addressing a distinct limitation of the other: the pyramid supplies multi-scale spatial coverage, while reconstruction provides the anomaly signal.

5.1.1 Prediction

The prediction method aims to determine abnormal events by comparing the actual data to expected data. The discrepancy between actual and expected can serve as the anomaly signal. This method is used with sequential data such as a video stream.

Techniques like Future Frame Prediction[13] and STAE[14] exhibit these properties. Future Frame Prediction uses consecutive sequential frames (I_1,I_2,...I_t), the model learns to predict the next frame (I_{t+1}). If the prediction closely matches the actual frame, the event is most likely normal. If there is a large discrepancy, it is marked as abnormal. STAE takes a slightly different approach where it progressively assigns lesser importance to distant frames and focuses the model on learning motion trajectories of existing objects rather than unpredictable appearances of new objects entering the scene.

5.1.2 Reconstruction

The underlying philosophy of reconstruction-based methods is that an image can be decomposed into an intermediary representation that is trained exclusively on known normal patterns. At inference time, the reconstructed image would have its anomalies suppressed since the image was compressed into an intermediate representation that can only represent normal patterns. Comparing the difference between the input and output images/features produces a reconstruction error that may serve as an anomaly signal.

Principal Component Analysis (PCA) and Variational Auto Encoders (VAE) methods leverage a bottleneck to compress the input into a lower dimensional representation. Since the bottleneck is trained exclusively on normal data, it learns to only preserve the most prominent normal structures. Anomalous features, lacking representation in this learned subspace, are effectively discarded during compression. The reconstructed image is therefore constrained towards normality, and the anomaly score is the difference between the input and reconstructed output. Diffusion relies on introducing noise to obfuscate and corrupt the input. If the anomaly is sufficiently obfuscated by the noise, the denoising process will reconstruct the image as normal since the model only learned to denoise normal patterns. The anomaly score then is the difference between the input and the reconstructed output.

Image Reconstruction Image reconstruction methods sub-category compares images at the image level. DRÆM[15] is capable of handling various anomalies as it was trained with artificial anomalies. Using the encoder-decoder architecture, it reconstructs patterns in the image to share a similar distribution to normal samples. The loss function encourages both pixel accuracy with L2 distance and structural accuracy by inclusion of a structural similarity index measure. Similarly, AE-SSIM[20] uses structural similarity index as well, however the key distinction is that AE-SSIM forgoes any pixel level loss metric as the authors believes it to be inaccurate around edges. Additionally, DRÆM jointly trains a segmentation network to predict anomaly maps from the input and reconstructed images. RIAD[24] takes a different approach, where it randomly removes partial image regions and uses inpainting to reconstruct. During training, loss is similar, where it includes SSIM and L2, but also introduces a loss for multi-scale gradient magnitude similarity to compare the intensity of pixel change. AnoGAN[21] uses 500 backprop steps through the GAN network with the actual image used to nudge the generated image to closely match the actual. If anomaly signal is determined by how close the generation matches the actual. MemAE[18] utilizes a memory-augmented autoencoder where images passed into the encoder are used to query a memory bank of learned normal patterns, these patterns are selected via cosine similarity thresholding and

merged into a single feature vector which the decoder generates a reconstructed image from.

Temporal Image Reconstruction Temporal image reconstruction introduces a temporal dimension. Instead of dealing with a single images, it deals with sequential images. Both STAE[14] and the Spatialtemporal Autoencoder[22] ingests n frames (16 for STAE, and 10 for Spatialtemporal AE) to reconstruct the same n frames at the same time steps which are used for comparison. The difference is that STAE also predicts 16 additional frames during its training process. Additionally STAE utilizes a 3D CNN autoencoder pipeline while SpatialTemporal AE uses a 2D CNN encoder and decoder with a ConvoLSTM autoencoder in the middle. 2D CNN AE[23] works similarly, but with its key distinction being the re-purposing of a dimension for the temporal context by stacking gray-scaled images. The comparisons are handled with STAE using squared L2 error, while Spatialtemporal AE and 2D CNN AE use L2 error.

Feature Space Reconstruction Image reconstruction methods sub-category compares images with the latent or feature vectors. Reverse Distillation [16] and MaskVIT[25] are both similar as they reconstruct from at the feature level. Reverse distillation uses a student decoder to reconstruct the teacher's feature representation from a compressed bottleneck embedding. While, MaskVIT uses a compressed feature representation and masked attention in its pipeline to reconstruct its feature map. Both methods use cosine dissimilarity metric to compare the original against the reconstructed features. GANomaly[19] being another GAN architecture similar to AnoGAN, but starts with the actual image, encodes and decodes, then encodes again. This feature vector being the result of the 2nd encoding step is compared against the image at the first encode step.

Temporal Feature Space Reconstruction Stacked RNN[17] uses the squared L2 distance for comparison between the actual and reconstructed frames in its CNN feature space. The method incorporates context from prior frames and learned patterns during its reconstruction process.

5.1.3 One Class Classification

One-class classification methods involve learning a representation of a single normal homogeneous class during training, such that classifications falling outside the normal class decision boundary or in low-density regions of the distribution are flagged as anomalous.

Boundary Boundary based one class classification solutions like CD-SVDD[31] learns the minimum-volume hypersphere to enclose normal data, classifying samples outside the boundary as anomalous.

Distribution Explicit density estimation methods for one-class anomaly detection model the probability density of normal data or its learned representations. PaDiM[28] is one such example that decomposes an image into a spatial grid to fits a multivariate Gaussian per patch position within its pipeline. Similarly, CutPaste[29] uses a Gaussian density estimator on its features to build a distribution. Building upon the Gaussian distribution ideas, FastFlow[30] uses normalizing flows to learn an invertible mapping from normal image features to a Gaussian distribution. Alternatively, GEOM[32] input images are scored by how likely their softmax patterns fall under Dirichlet distributions which has been fitted to normal data. AnoGAN[21] is trained exclusively on healthy data; the anomaly score measures how well a query fits the implicitly learned normal distribution. The discrimination loss checks whether the generated image lies on the learned manifold.

Distance Distanced based one class solutions evaluate the a distance measure from how far a sample's features are from normality. PANDA[33], SPADE[27] and PatchCore[26] all utilize nearest-neighbor based scoring. The difference resides in the data being compared. PANDAs compares global image features, while SPADE focuses on local features comparisons. PatchCore evaluates the patch features against the normal patch features stored in its memory bank. The anomaly signal is produced by the maximum distance scores, with the larger scores differing significantly from normality. CSI[34] also falls under this categorization, but uses a hybrid approach that combines cosine similarity to the nearest training sample with feature magnitude and shift-classification scores. Reverse Distillation [16] also falls under distance-based scoring, measuring cosine distance between corresponding features from a frozen pre-trained teacher and a student trained on normal data. These discrepancies are combined across multiple scales to identify anomalous regions. For GANomaly[19], rather than comparing input and reconstructed images at the pixel level, a second encoder re-encodes the reconstructed image back into latent space, and the anomaly score is the distance between the two latent vectors.

5.1.4 Multiclass Classification

Multiclass classification methods frame anomaly detection as the task of learning discriminative representations across multiple distinct object or texture categories simultaneously, rather than training a separate model per class. A single unified model is trained to classify inputs into one of several known normal categories, and anomaly scoring is derived from classification uncertainty or feature-space distances. The key advantage is scalability: a single model can serve as the anomaly detector for many product types and can output the type of anomaly.

GEOM[32] distinguishes among each of the 72 transformations with a single CNN model, the output from this level generates 72 probability vectors. The end result is unified into one final score, however this falls under Multiclass due to its intermediary step.

5.1.5 Representation/Pre-trained Models

This involves the utilization of a pre-trained network which has been trained on a large dataset such as ImageNet and has developed a rich hierarchical feature representations of the related task. By applying transfer learning, the weights of the pre-trained backbone are frozen, thereby preserving the learned representations, upon which additional components such as adapters, projection heads, and task-specific layers can be built and trained. This allows the model to be adapted to downstream tasks while preventing catastrophic forgetting. Alternatively, fine-tuning unfreezes some or all layers of the pre-trained backbone, allowing the weights to be updated during training. This enables the model to further adapt its representation to the target domain while leveraging the rich features learned during pretraining.

Image Net Family A majority of methods used pre-trained networks trained on ImageNet dataset, for the following we will consider the models apart of the ImageNet family of models. SPADE[27] uses WideResNet-50x2 purely as a feature extractor. While PANDA[33] takes it a step further and applies additional finetuning on ResNet's 3rd and 4th layers. Similarly, CutPaste[29] applies finetuning ontop of a EfficientNet-B4 backbone. Models like FastFlow[30] follows the same pattern of frozen backbone plus a trained adapter. Uniquely, Uninformed Students[38] teacher is built by distilling information from a frozen ResNet-18 model. In the same vein, Reverse Distillation[16] selects a backbone from the ResNet models in its teacher-student pipeline. Few

other papers such PatchCore[26] and PaDiM[28] as take this approach of evaluation across multiple backbones.

DINO Family Alternatively to ImageNet, the DINO family of models are another popular choice to provide the pipeline with generalized knowledge. MaskVIT [25] and AnomalyDINO[37] are two methods that incorporate these models. MaskViT introduces a frozen 12-layer pre-trained DINO vision transformer as its encoder. AnomalyDINO leverages DINOv2 as its frozen backbone.

Language Model Family The Large Language Model family of pre-trained models involves the usage of LLMs as its backbone. AnomalyGPT[35] and IAD-GPT [36]'s generalized methodology treats a frozen ImageBind-Huge, which handles feature encoding of imagery, is projected onto a frozen Vicuna-7B language model. The primary difference is how each methods processes the anomaly. While AnomalyGPT uses a direct approach of a linear layer to connect the image encoder to the language model, IAD-GPT addresses the issue with a Text-Guided Enhancer which increases relevancy towards image features with text associations.

Others These methods involve unique backbone models. Stacked RNN[17] utilizes pre-trained CNN as frozen feature extractor trained on the UCF101 dataset to extract features in each frame of a video. MIL[39] extracts the FC6 layer from a frozen C3D backones which handles the 16-frame forward pass for feature extraction. Future Frame Prediction[13] utilizes a frozen pre-trained FlowNet to steer the model towards predictions with consistent motion and realistic predictions.

5.1.6 Memory Bank

Rather than encoding normality implicitly into the model weights during training, normal patterns are stored in an explicit, structured data structure that is separate from the core network parameters. This memory serves as a reference bank of normal features against which test samples are compared. The memory may be populated at inference time or during training. Anomalies are identified as samples that deviate significantly from the stored representations, with the anomaly signal derived from the relationship between the test sample and the memory contents rather than from the model's internal transformations.

Nominal Feature Referencing This category addresses features stored in a memory bank for direct referecing by the model within its pipeline. SPADE[27] stores normal features vector describing the entire image in its memory bank. Test samples are then compared against via nearest-neighbor distance and a subsequent does a local feature extraction and comparision at various feature map resolutions. Methods like PatchCore[26] and AnomalyDINO[37] take a different approach and store only nominal patch-level features. PatchCore memory optimization introduces coreset selection to reduce memory bank usage by up to 100x while retaining nearly identical performance by selecting a small representative subset of patch features that represent a larger group of patches. The idea is to maximizes coverage of the feature space by iteratively picking points that are farthest from any already-selected point. AnomalyDINO's approach is more direct, as it stores patch level features directly, however they do apply augmentations to increase hits. The method fetches for closest matches via cosine distance. For language model based approaches, Both AnomalyGPT[35] and IAD-GPT[36], in the few-shot mode, maintains four memory banks that stores information at various depths in the network.

Learned Memory Vectors MemAE[18] takes a different approach, as its memory banks is jointed learned at training time. During training, between the encoder and decoder, MemAE implemented a memory module M to retrieve the most relevant memory via an attention-based addressing operator.

5.1.7 Data Augmentation

Anomaly detection can be augmented by synthetically generating pseudo-anomalies during training. Since real anomalous samples are scarce or unavailable, artificial anomalies are synthesized by applying perturbations to normal images, such as cutting and pasting random patches, applying noise, or blending textures from external sources. The model is then trained to discriminate between normal and synthetically anomalous samples, learning a decision boundary that generalizes to real anomalies at inference time. Simple rotations, flips, shifts without meaningful purpose do not count.

Transformation-Based Augmentation This sub-category consists of simple transformations such as rotations, shifts and such. AnomalyDino[37] is an example of such method as the authors adds rotational and masking augmentations to diversity features stored in its memory bank. GEOM[32] is another example where the entire method is built on 72 geometric transformations that get fed into the network. CSI [34] takes a different approach as the method synthesizes pseudo-anomalies by applying transformations that maximally shift the input distribution, then trains the model to discriminate normal from shifted samples.

Synthetic Anomaly Generation Synthetic anomaly generation is a more sophisticated approach which introduces fake anomalies to aid in generalization of the model. DRÆM[15] is a method which follows this philosophy as anomalies are generated via perlin noise masks that are blended with textures from the Describable Texture Dataset. Similarly, CutPaste[29]'s strategy involves cutting patches from an image and pasting them at random locations of the image. AnomalyGPT [35] and IAD-GPT [36] expand on this by introducing Natural Synthetic Anomalies technique (NSA) which starts with cutting a pasting patches like the CutPaste technique, but then introduces poisson blending when pasting the patches to better blend into its surrounding region.

Corruption-Based Reconstruction RIAD[24] introduces anomalies by dividing the image into grid based groups for which random groups are zeroed out thereby creating a self-supervised inpainting objective without needing any external anomaly samples or textures.

5.1.8 Pyramid

Pyramid based methods leverage feature representations at multiple levels to capture anomalies that manifest at various spatial or feature resolutions. The core concept behind each layer in the pyramid is such that earlier layers capture fine-grained details such as lines, foundational geometry, local textures which become progressively more semantic and categorical deeper into the network. By jointly evaluating multiple layers of the pyramid, textural anomalies, such as localized defects, scratches and cracks are captured alongside functional anomalies such as missing components.

Multi-Level Features Multi-level features are feature vectors extracted at multiple levels of the network. SPADE[27] is a straightforward example as it concatenates features from various depths of the ResNet blocks to capture varying spatial features. This is used for to cross-reference

against nominal features. PaDiM[28] also concatenates features at various layers, but it is to fit a multivariate Gaussian distribution. PatchCore[26] handles it differently as they only concatenate two intermediate feature hierarchies to be used in conjunction with its memory bank. FastFlow[30] only utilizes three separate levels in the ResNet network which are independently processed then averaged. As AnomalyGPT[35] and IAD-GPT[36] are closely related in methodologies, they both extract features from layers 8, 16, 24, 32 of ImageBind. Reverse Distillation[16] extracts features from the frozen teacher's feature maps to be concatenated and bottle-necked before forwarding to the student network.

Multi-Scale Receptive Fields Uninformed Students[38] Multiple student–teacher ensemble pairs are trained with varying receptive field sizes. Each pair captures anomalies at a single scale determined by its receptive field, and the normalized anomaly scores are averaged across pairs to detect anomalies at multiple spatial resolutions. Stacked RNN[17] gradually partitions the feature map into fine regions such as 1x1, 2x2, and 4x4, then a max pool aggregates them into a single feature vector.

5.1.9 Vision Language

Vision-Language methods are constructed from multi-modal models such as CLIP that jointly embed images and text into a shared latent space for semantic representation, enabling anomaly detection via natural language. Rather than training on task-specific imagery, these methods leverage the model's broad semantic understanding acquired during large-scale vision-language pretraining, using text prompts to describe normal and anomalous conditions. Due to the model's inherent fundamental understanding and generalization capabilities derived from the combination of language and vision, this paradigm enables zero-shot and few-shot anomaly detection, where zero-shot refers to inference without any examples from the target domain; thereby foregoing or reducing reliance on domain-specific imagery. At inference time, image regions whose features align more closely with anomalous text descriptions rather than with normal descriptions are flagged as anomalous. Furthermore, outputs may produce textual explanations with reasoning to improve perceived interpretability.

AnomalyGPT[35] embeds both images and normal/abnormal text descriptors into a shared space via the decoder's feature-text matching, and leverages the LLM to produce natural language anomaly judgments, enabling few-shot transfer to unseen object categories through in-context learning with text prompts rather than retraining. While IAD-GPT[36] uses GPT to generate category-specific anomaly text prompts (e.g., cracks, tears for leather) that activate CLIP's segmentation, replacing generic normal/abnormal templates.

5.1.10 Ensemble

Ensemble methods aggregate anomaly scores from multiple independently trained models to produce a more robust and stable detection signal. Rather than relying on a single model's output, several models, potentially trained with different random seeds, augmentation strategies, or architectural configurations, each produce an anomaly score that is combined through averaging, voting, or other aggregation functions. The resulting score benefits from variance reduction across individual models, smoothing out per-model sensitivity to initialization or data sampling. Ensembling is agnostic to the underlying methodology; any combination of reconstruction, one-class classification, or augmentation-based approaches may be aggregated, making it a general-purpose scoring strategy that operates downstream of the core anomaly modeling pipeline.

Independently Trained Models in an Ensemble CutPaste[29] handles ensembling in a straightforward fashion where 5 separate models of the same architecture are independently trained from random seeds and their anomaly scores are combined. Uninformed Students[38] creates an ensemble of $M \geq 1$ student CNN networks for the purposes of feature prediction in the student portion of the model's pipeline. Students are randomly initialized and each pair specializes in a different receptive field (kernel size). As an example, one student teacher pair may focus on regions of 17x17 while another may have 33x33. Ensemble outputs are combine via equal weighting from the predictive distribution at each pixel.

Single Train Model Inferenced with Various Parameters RIAD[24] addresses ensembling sightly differently as it trains a single model. At inference, the model is ran for various masking sizes (e.g. 2x2, 4x4, 8x8 pixels). When aggregating, an average is applied to each pixel's anomaly score in the anomaly map. CSI[34] takes a similar approach by training a single model. The same model is ran multiple times with various augmentations applied to the input image for each run. These augmentation are random such as cropping, flipping, and color manipulations. The resulting normality score for each run are averaged.

5.1.11 Teacher Student

Teacher-student methods exploit the knowledge distillation framework for anomaly detection. A pre-trained teacher network, which encodes rich feature representations, is paired with a student network that is trained to mimic the teacher's output on normal data. Because the student is only exposed to normal samples during training, it learns to reproduce the teacher's representations for normal inputs but is expected to fail for anomalous inputs, which lie outside the student's learned mapping. The discrepancy between the teacher's and student's feature representations at inference time serves as the anomaly signal.

Uninformed Students[38] uses a frozen teacher from a pre-trained network where it is used to train the ensemble of student CNN networks to predict features. The each student in the ensemble is combined via predictive variance and a separate comparison branch handles the teacher vs students via L2 distance. Scores from both branches are combined to produce a pixel level anomaly map. Similarly, Reverse Distillation[16] uses a frozen teacher network as well during inference and training. It is used to train a single student decoder to reconstruct features.

5.1.12 Continual Learning

Continual learning, also referred to as incremental or lifelong learning, addresses the scenario where the anomaly detection model must adapt to new object categories, domains, or defect types over time without forgetting previously learned normal representations. In industrial settings, new product lines are introduced continuously, and retraining from scratch on all accumulated data is computationally prohibitive. Continual anomaly detection methods aim to incrementally update the model as new classes arrive while preserving detection performance on previously seen categories.

For both PANDA[33] and PatchCoreCL[40], the incoming data used to update the model consists of images known to be normal. PANDA uses a continual learning inspired regularization method, elastic weight consolidation, during training to preserve important features. PANDA is not truly continual learning. However, PatchCoreCL implements continual learning by sequentially processing new task and updating its task-specific memory banks while mitigating forgetfulness of prior learned tasks. This is done by keeping the total memory constant by equally re-balancing

all per-task banks via coreset subsampling each time a new task arrives, since the frozen backbone makes weight-based CL techniques inapplicable.

5.2 Training Paradigm

Before any of the above techniques can produce useful results, they need to be trained. This section covers the training paradigms used to train the architecture at a high level. The common paradigms involved in vision anomaly detection involves unsupervised, self-supervised, and supervised. It is important to note that terminologies in anomaly detection are not aligned with commonly accepted definition in machine learning. **Unsupervised** trains on normal data, **Self-supervised** uses normal data but generates its own training signal, and **Supervised** approaches may require partial labeling to distinguish between normal and abnormal data. None of the reviewed works utilize reinforcement learning. The complete set of techniques and their associated training paradigm can be found in Table 2.

5.2.1 Unsupervised

Unsupervised methods operate under no availability of labeled data. The method learns patterns exclusively from all provided data which is assumed to be purely normal. This would typically model the distribution of normal images. During inference, samples that deviate significantly from the learned distribution are flagged as anomalous. This approach is especially valuable when anomalies are rare and unknown in advance, making it impractical to collect labeled examples.

5.2.2 Self-Supervised

Self-supervised is not conventionally acknowledged as a distinct category and is often classified as an unsupervised technique in the anomaly detection field [41][42]. Such is the case of DRÆM [15] which the authors classify as unsupervised. Self-supervised methods generate their own training signal by creating artificial tasks from unlabeled normal data. For example, applying known alterations to an image and training the model to identify or reverse those alterations. This exposes the model to both the synthetic data alongside the generated labels.

5.2.3 Supervised (Weakly or Fully)

Supervised methods utilize labeled data to guide training, where annotations indicate the presence or classification of anomalies. In a fully supervised setting, labels are provided at the same granularity as the prediction target. In a weakly supervised setting, labels or a mixture of incomplete labels are provided at a coarser granularity than the prediction target, requiring the model to infer finer-grained predictions from broader supervision signals.

5.3 Zero-Shot/ Training Free

Training free has no training loop involved for the core techniques. Technically these models usually contain a portion of the model that is pre-trained on another training paradigm, but since the core technique leverages this pre-trained network and does no additional training of its own, it will fall under this categorization. Zero-shot inference extends this further, where the model is applied directly to a target domain or anomaly type it has never explicitly been exposed to, relying on the generalization capacity of the pre-trained backbone.

5.4 Training Tactics

This section highlights some standout considerations and novel ideas applied for the training process. The following will show an applied approach to extend the functionality of ideas from other subdomains of machine learning to different subdomains where applicable. Such as a task specific adaptation of Structural Similarity Index Measure which was not initially intended for machine learning, but has now been widely used in machine learning. PANDA[33] also borrows ideas from other machine learning fields and applies them directly in the training process to be a form of regularization. Therefore it is encouraged that techniques from other fields can be adapted for the purposes of anomaly detection. Papers suggest that widely used machine learning methods still apply here as well from the machine learning field in general.

5.4.1 Knowledge Preservation and Regularization

While training is straight forward for most techniques, there are some standout practices which improves model performance. The following will highlight these practices. PANDA[33] for example, uses a compactness loss to migrate the image's feature vectors towards a fixed center thereby compacting them. Initially, anomalous (unseen normal) samples will get pulled less towards the center, but then eventually proceed to collapse. This is why the regularization technique of PANDA-Early, PANDA-SES, and PANDA-EWC was introduced. PANDA-Early is a common approach of stopping at a fixed number of iterations and PANDA-EWC is elastic weight consolidation which is borrowed from continual learning. It penalizes weights that when a small change is applied produces a big impact in the network's error. This change is to counterbalance the compactness loss function by trying to mitigate drastic changes in the network but also support the clustering of normal features. While PANDA-SES is a novel approach that is developed for compactness loss. Periodic checkpoints are created and each checkpoint is scored via the ratio of distance of a test image from the center over the average training distance from the center. It takes the checkpoint with the highest score.

5.4.2 Addressing Loss Function

Another strategy to steer the model during training is to redefine what we optimize for by adjusting loss function. AE (SSIM)[20] is one such example that addresses the loss function directly by introducing the Structural Similarity Index Measure (SSIM) metric into the loss. Since AE (SSIM) falls under the category of reconstruction, the training focuses on discourage large variations in luminance, contrast, and structure between the reconstructed image and the original input image.

5.4.3 Temporal Considerations

Pertaining to image sequences (videos). According to Zhao et al.[22], It is important that the time between each frame stays consistent. This is because a change in the velocity of an object caused by non-constant sampling of frames would be considered an anomaly. Furthermore, their model, Spatio-Temporal AutoEncoder[22], introduces a novel concept by decreasing prediction loss weights for frames further into the future. Meaning that prediction of the next frame is considered more than prediction of a frame that is sixteen steps into the future.

5.4.4 Staged Optimization

As demonstrated by IAD-GPT [36], staged optimization is where different parts of the model can be trained in different phases. Each phase can include a different loss functions and which

weights remain frozen or active. This allows training to be more stable while reducing undesired cross-module interference.

6 Datasets

This section covers datasets that have been used as benchmarks or training data for anomaly detection methods. Each paper may have different interpretation of optimal use of the dataset and may only utilize a subset of the full set.

Table 3: Table demonstrates the relationship of usage between vision datasets and papers/methodologies that utilized the datasets
[1] Dataset created by the paper's authors or co-authors
[2] Papers that are not part of the seminal works
[3] Dataset no longer can be found, an approximation based on paper's information is conducted

Dataset	Total	Type	Annotation	Domain	Description	Used By
MVTecAD[43]	5,354 images	Image	Labels + pixel-level	Industrial	Manufacturing Defects	[37] [26] [27] [28] [15] [29] [30] [16] [38][1] [33] [24] [25][2] [36][2] [35][2]
VisA[44]	10,821 images	Image	Labels + pixel-level	Industrial	Manufacturing Defects	[37] [25][2] [35][2]
MTD[45]	1,344 images	Image	Labels + pixel-level	Industrial	Manufacturing Defects	[26]
MNIST[46]	70,000 images	Image	Labels	General	Numerical Digits	[29] [16] [18] [19] [38] [32]
CUHK Avenue[47]	30,651 frames	Video	Bounding box	Social Sciences	People Surveillance	[17] [18] [39] [13] [22] [23] [14] [24]
UCSD Peds 1 & 2[48]	18,560 frames	Video	Frame-level labels + partial pixel masks	Social Sciences	People Surveillance	[17] [18] [39] [13] [22] [23] [14] [24]
UBA[49]	230,275 images/patches	Image	Labels	Social Sciences	Security X-ray	[19][1]
Woven Fabric[20]	250 images	Image	Labels	Industrial	Textile Defect	[20][1]
SD-OCT[21]	290 volumes	Volumetric	Labels	Medical	Optical Coherence Tomography	[21][1]
UCF-Crime[39]	1,377,653 frames	Video	Labels	Social Sciences	Crime Surveillance	[39][1]
Toy Pedestrian[13]	1,452 frames	Video	Labels	Social Sciences	Synthetic People	[13][1]
Traffic[14]	327,889 frames	Video	Labels	Social Sciences	Traffic Surveillance	[14][1]
STC[50]	317,398 frames	Video	Frame-level labels	Social Sciences	People Surveillance	[26] [27] [28] [17][1] [18] [13]
DTD[51]	5,640 images	Image	Labels	Physical Sciences	Textures	[15] [34]
ImageNet[52]	1,431,167 images	Image	Labels + bounding boxes	General	Everyday Objects	[15] [29] [38] [34]
DAGM[53]	16,100 images	Image	Labels	Industrial	Surface Defects	[15]
CIFAR-10/100[54]	60,000 each	Image	Labels	General	Everyday Objects	[29] [30] [16] [31] [18] [19] [38] [32] [34] [33]
dSprites[55]	737,280 images	Image	Factor labels	General	Synthetic	[29]

Table 3: Table demonstrates the relationship of usage between vision datasets and papers/methodologies that utilized the datasets
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[3] Dataset no longer can be found, an approximation based on paper's information is conducted

Dataset	**Total**	**Type**	**Annotation**	**Domain**	**Description**	**Used By**
BTAD[56]	2,540 images	Image	Labels + pixel-level	Industrial	Manufacturing Defects	[30]
FashionMNIST[57]	70,000 images	Image	Labels	Social Sciences	Fashion	[16] [32] [33]
Caltech 256[58]	30,607 images	Image	Labels	General	Objects	[16] [34]
MovingMNIST[59]	200,000 frames	Video	None	General	Moving Digits Synthetic	[17]
NanoTWICE[60]	40 images	Image	Pixel-level	Industrial	Nano Material Defects	[20]
FFOB[61]	72,352 images	Image	Labels	Social Sciences	Security X-ray	[19]
UMN[62]	7,739 frames	Video	Labels	Social Sciences	Crowd Surveillance	[39]
BOSS[63][3]	∼48,600 frames	Video	Labels	Social Sciences	People Surveillance	[39]
ASIRRA[64]	25,000 images	Image	Labels	General	Animals	[32]
Subway[65]	186,650 frames	Video	Event-level labels	Social Sciences	People Surveillance	[39] [22] [23]
Abnormal Crowd[66]	44,091 frames	Video	Labels	Social Sciences	Crowd Surveillance	[39]
SVHN[67]	99,289 images	Image	Labels	General	Digits	[34]
LSUN[68]	72,417,879 images	Image	Labels	General	Scenes	[34]
CUB-200[69]	11,788 images	Image	Labels + bounding boxes + parts	General	Birds	[34] [33]
Stanford Dogs[70]	20,580 images	Image	Labels + bounding boxes	General	Animals	[34]
Oxford-IIIT Pet[71]	7,390 images	Image	Labels + pixel-level + ROI	General	Animals	[34]
Oxford 102 Flower[72]	8,189 images	Image	Labels + pixel-level	Life Sciences	Flower	[34] [33]
Food-101[73]	101,000 images	Image	Labels	Life Sciences	Food	[34]
Places-365[74]	2,168,460 images	Image	Labels	General	Scenes	[34]
TILDA[75]	3,200 images	Image	Labels	Industrial	Textile	[34]
WBC	484 images	Image	Labels	Health Sciences	Medical Microscopy	[33]
DIOR[76]	23,463 images	Image	Bounding boxes	Physical Sciences	Remote sensing	[33]
BMAD[77]	81,056 images/slices	Image	Labels + pixel-level (3 subsets)	Health Sciences	Medical Scans	[40][2]

6.1 Dataset Descriptions

MVTecAD[43] MVTec anomaly detection dataset contains 5,354 high-resolution color images of ten objects and five textures with 73 different types of defects mimicking real-world industrial inspection scenarios: carpet, leather, bottles, pills, screws, transistors, etc. Dataset also contains pixel-accurate ground truth regions for 1888 images. This dataset may be used for classification and segmentation outputs. [43]

VisA[44] Comprised of 10,821 images total split into 9,621 normal and 1,200 anomalous samples with 12 different objects and anomaly subclassification. Classes include objects such as PCBs, candles, capsules, cashews, etc. [44]

MTD[45] Magnetic tile defects dataset contains a total of 1,344 images with pixel level labeling for segmentation and defect classification such as crack, frays, break, uneven, etc. [45]

MNIST[46] Modified national institute of standards and technology database is a collection of handwritten digits from zero to nine, represented in 70,000 gray scales images that are 28 by 28 pixels. [46]

CUHK Avenue[47] Chinese University of Hong Kong avenue dataset is comprised of 16 training and 21 testing video clips captured at the Chinese University of Hong Kong campus avenue. 30,652 total frames split into 15,328 training and 15,323 testing. Abnormal events are labeled with rectangular bounding boxes.[47]

UCSD Peds 1 & Peds 2[48] University of California San Diego pedestrian contains Peds 1 & Peds 2. Peds 1 contains 34 training and 36 testing video samples of groups of people walking towards and away from the camera at a top down view. Peds 2 contains 16 training and 12 testing video samples of pedestrian movement parallel to the camera plane. Each clip has corresponding ground truth annotations as an anomaly binary flag per frame. Pixel level binary masks are only present in a subset.[48]

UBA[49] University baggage anomaly dataset is a private dataset due to security reasons. This is a sliding window patched-based dataset comprised of 230,275 image patches derived from baggage X-ray imagery. 122,803 abnormal classes split among three classifications knife, gun, and gun components. [19]

Woven Fabric Texture Dataset[20] The dataset pertains to close up thread patterns of fabrics. The dataset contains 200 defect-free imagery split between two woven fabric texture types. With an additional 50 images containing various defects. Images are gray scaled of size 512 by 512 pixels. [20]

SD-OCT[21] This data set related to spectral-domain optical coherence tomography retinal volumes. There are a total of 290 clinical OCT volumes of resolution 496 by 512 by 49 voxel with 49 representing B-scans (image slices). 20 of the 290 are for testing with 50/50 split of healthy and pathological cases. [21]

UCF-Crime Dataset[39] This dataset includes 13 real world anomalies such as accidents, burglaries, explosions, fighting, etc in a total of 1900 videos equivalent to a runtime of 128 hours. This dataset was annotated by ten trained computer science vision experts of varying levels. The data was collected from Youtube and LiveLeak. [39]

Toy Pedestrian[13] A 2D oversimplified dataset of solid color stick figures and solid color vehicles on a white backdrop containing a total of 1452 frames split between 210 training frames and 1242 testing frames.[13]

Traffic Dataset[14] Collected from real-world traffic surveillance systems. Five different scenarios with each containing 30 minutes of normal and 10 minutes of testing footage. [14]

STC[50] Shanghai Tech campus dataset contains a total of 317,398 frames split into 274,515 training and 42,883 testing frames. There are 13 diverse scenes with 130 distinct abnormal events and approximately 5% of the total frames are abnormal frames. [50]

DTD[51] Describable textures dataset contains 5640 labeled images split among 47 categories depicting various textures like patterns, flowers, leaves, fabrics, etc. extracted from Google and Flickr. 120 images per category with image sizes from 300 by 300 to 640 by 640 pixels. Dataset contains a balanced split into train, validation, and test [51]

ImageNet[52] The most popular ImageNet subset contains a total of 1,431,167 images split into 1,281,167 training, 100,000 testing and 50,000 validation of general, generic, non-specific, images for the purposes of image classification and localization.[52]

DAGM[53] Deutsche Arbeitsgemeinschaft für Mustererkennung dataset contains ten classes of industrial surface defect high-resolution gray scale images with a mixture of defects and normal cases. There is a grand total of 16,100 images, but split between a development and competition dataset. 14,000 non-defective and 2,100 defective images. Each class is a 50/50 balanced split. Defective images contain exactly one defect per image. [78]

CIFAR-10, CIFAR-100[54] The Canadian Institute For Advanced Research datasets are labeled subsets of the 80 million tiny images dataset created by Krizhevsky, Nair, and Hinton. Both contain 60,000 32 by 32 color images split into 50,000 training and 10,000 testing samples. CIFAR-10 organizes images into 10 mutually exclusive classes such as airplanes, automobiles, birds, cats, and ships, while CIFAR-100 expands to 100 fine-grained classes further grouped into 20 coarse superclasses, providing both a fine and coarse label per image.[54]

dSprites[55] Disentanglement testing sprites dataset is a synthetically generated dataset designed to evaluate how well unsupervised models recover underlying generative factors. It contains 737,280 black and white 64 by 64 images procedurally generated from every possible combination of 6 independent latent factors: shape (square, ellipse, heart), scale, orientation, and x and y position. [55]

BTAD[56] BeanTech anomaly detection dataset is a real-world industrial anomaly dataset released by beanTech and the University of Udine, Italy. It contains 2,540 RGB images across three unspecified industrial products with varying resolutions of 1600 by 1600, 600 by 600, and 800 by 600 pixels respectively. The training set consists only of defect-free normal images, while the test set contains a mixture of normal and abnormal images, supporting both anomaly detection and pixel-level localization via ground truth masks. [30]

FashionMNIST[57] Fashion-MNIST is a dataset of grayscale apparel article images intended as a more challenging drop-in replacement for the original MNIST dataset. It contains 70,000 images split into 60,000 training and 10,000 testing examples across 10 clothing categories including t-shirts, trousers, pullovers, dresses, coats, sandals, shirts, sneakers, bags, and ankle boots. Each image is 28 by 28 pixels, matching the exact image size and train/test split structure of the original MNIST dataset. [57]

Caltech 256[58] Caltech-256 is a challenging object category dataset containing 30,607 images spread across 256 categories. It is an expansion of the original Caltech-101, with the number of categories more than doubled, a minimum of 80 images guaranteed per category, and the addition of a dedicated clutter category for testing background rejection. Images were collected from Google Images and manually screened for category fit. The dataset supports image classification and background rejection tasks. [58]

MovingMNIST[59] Moving MNIST is a synthetic video prediction dataset consisting of 10,000 sequences each of length 20 frames depicting two handwritten digits moving and bouncing within a 64 by 64 pixel grayscale frame. Intended as a standardized test set for sequence prediction and reconstruction tasks. [59]

NanoTWICE[60] NanoTWICE is a dataset of scanning electron microscope (SEM) images of electrospun nanofibrous materials intended for anomaly detection and defect localization. The dataset contains 40 test images of nanofibrous material surfaces, accompanied by a pre-trained sparse coding dictionary and anomaly detection density model. Defects manifest as structural irregularities in the fiber patterns. The dataset was introduced alongside a MATLAB GUI tool for pixel-level anomaly detection via sparse coding. [60]

FFOB[61] The Full Firearm versus Operational Benign dataset is a UK government evaluation dataset comprising expertly concealed firearm threat items and operational benign non-threat imagery sourced from commercial X-ray security screening operations of baggage and parcels. The dataset contains 4,680 full firearm images as the abnormal class and 67,672 operational benign images as the normal class, with an 80/20 train/test split applied to normal samples. [19]

UMN[62] The University of Minnesota unusual crowd activity dataset consists of five different staged videos where people walk around and after some time start running in different directions. The anomaly present in the dataset is characterized solely by the running action. Each video averages 1,290 frames with a total dataset length of approximately five minutes. [39]

BOSS[63] The Behavior Observation for Safety and Security dataset was originally captured as part of the Eureka Celtic Initiative "On Board Wireless Secured Video Surveillance" project. It consists of video and audio recordings of acted actions inside a moving train using multiple

calibrated cameras. The dataset contains 12 videos averaging 4,052 frames for a total runtime of approximately 27 minutes. Anomalies are performed by actors and include behaviors such as harassment, panic situations, fainting, fighting, and a person with a disease, alongside normal activities. It supports pose, action, and interaction recognition tasks. [63]

ASIRRA (Cats VS Dogs)[64] Animal Image Recognition for Restricting Access (ASIRRA) dataset originates from a CAPTCHA system originally deployed by Microsoft Research in partnership with Petfinder.com. The full image pool is sourced from Petfinder.com consists of approximately three million color images of cats and dogs are sourced from real animal shelter listings. The commonly used Kaggle subset contains 25,000 labeled images evenly split between 12,500 cats and 12,500 dogs for binary classification. Images vary in resolution but on average are 360 by 400 pixels, background, pose, and lighting, reflecting real-world photographic diversity rather than controlled capture conditions. [64]

Subway Entrance & Exit[65] Subway dataset contains two indoor surveillance video clips captured at a subway station divided into an Entrance and Exit sequence with a combined runtime of approximately 3 hours. The Entrance clip contains 121,749 frames and the Exit clip contains 64,901 frames. Anomalous events include pedestrians walking in the wrong direction and skipping payment. Annotations are provided as event-level labels rather than pixel-level masks and are evaluated on a per-event basis. [39]

Abnormal Crowd Dataset[66] Abnormal Crowd dataset contains 31 videos of crowded scenes with a combined runtime of approximately 24 minutes and an average of 1408 frames per video. Anomalous event categories include panic, fighting, congestion, obstacle, and neutral behaviors. [66]

SVHN[67] The Street View House Numbers dataset contains 99,289 real-world digit images cropped from Google Street View imagery, split into 73,257 training and 26,032 testing samples at 32 by 32 pixels in color, with an additional 531,131 less difficult extra training samples also available. Digits are labeled zero through nine and may appear alongside distracting digits in the periphery of each crop. [67]

LSUN[68] The Large Scale Scene Understanding dataset contains approximately 10 million labeled images across ten scene categories: bedroom, bridge, church outdoor, classroom, conference room, dining room, kitchen, living room, restaurant, and tower. Approximately 59 million labeled images across twenty object categories. Images are variable in resolution, distributed and released with the shorter dimension resized to 256 pixels. The dataset is intended for scene classification and object detection benchmarks. [68]

CUB-200[69] The Caltech-UCSD Birds 200-2011 dataset contains 11,788 images of 200 North American bird species split into 5,994 training and 5,794 testing images. Each image is annotated with bounding boxes, part locations, and attribute labels, making it a standard fine-grained visual recognition benchmark. [34]

Stanford Dogs[70] The Stanford Dogs dataset contains 20,580 images of 120 dog breeds sourced from ImageNet, split into 12,000 training and 8,580 testing images. The dataset is intended for

fine-grained image categorization and features significant intra-class variation due to pose, lighting, and background diversity. [34]

Oxford-IIIT Pet[71] The Oxford-IIIT Pet dataset contains approximately 7,390 images across 37 cat and dog breed categories with roughly 200 images per class. Each image is annotated with a breed label, head region of interest (ROI), and pixel-level trimap segmentation mask, supporting both classification and segmentation tasks. [71]

Oxford 102 Flower[72] The Oxford 102 Category Flower dataset contains 8,189 images of flowers common to the United Kingdom split across 102 categories with between 40 and 258 images per class. Images exhibit large variation in scale, pose, and lighting and are accompanied by segmentation masks and class label annotations. [34]

Food-101[73] The Food-101 dataset contains 101,000 images evenly distributed across 101 food categories with 750 training and 250 manually verified testing images per class. Images were collected from foodspotting.com and intentionally include some amount of noise in the training split to reflect realistic web-scraped conditions. [34]

Places-365[74] The Places365 dataset is a large-scale scene-centric dataset containing approximately 1.8 million training images and 36,500 validation images across 365 scene categories such as airport, beach, forest, and shopping mall. Images are sourced from the broader Places database of over 10 million images and are intended for scene recognition and environmental context tasks. [34]

TILDA Textile Texture[75] The Textile Texture dataset (TILDA) is a reference dataset for evaluating visual inspection methods on textile surfaces. It contains 3,200 grayscale images of eight textile surface types at a resolution of 768 by 512 pixels. For each textile type, eight classes are defined, one defect-free reference class, and seven defect classes. Each contains 50 images per class. [75]

WBC[33] The white blood cell dataset is a microscopy image collection comprised of 4 classes with 59 training and 62 testing images per class. The dataset depicts various white blood cell types at high magnification and is used to evaluate anomaly detection performance on out-of-domain medical imagery distinct from typical natural image datasets. [33]

DIOR[76] The dataset for object Detection in Optical Remote sensing images is a large-scale benchmark for object detection in optical remote sensing imagery. The dataset contains 23,463 images and 192,472 instances covering 20 object classes, and features a large range of object size variations in terms of both spatial resolution and inter- and intra-class size variability. Images are obtained under varying imaging conditions, weathers, seasons, and image quality, resulting in high inter-class similarity and intra-class diversity. [76]

BMAD[77] Benchmarks for Medical Anomaly Detection is a comprehensive evaluation benchmark consisting of six reorganized datasets from five medical imaging domains (brain MRI, liver CT, retinal OCT, chest X-ray, and digital histopathology). Contains a total of 81,056 images and slices sourced from BraTS2021, BTCV, LiTs, RESC, OCT2017, RSNA, and Camelyon16. Three datasets support pixel-level anomaly localization with segmentation masks, while the remaining

three provide sample-level image labels. Benchmark also integrates fifteen state-of-the-art anomaly detection algorithms and three evaluation metrics (Image AUROC, Pixel AUROC, and PRO). [77]

7 Discussion and Observations

This section sheds light on the broader challenges in anomaly detection, spanning the taxonomy of anomaly types, the impact of data quality and quantity on model development, the assumptions that underpin common detection approaches and their failure modes, and the various forms of model drift that arise when deployment conditions diverge from training conditions.

7.1 Anomaly Types

Anomalies may manifest across different aspects of image and video data, and can be broadly categorized by their occurrence pattern, extent, appearance, and semantic meaning. At the occurrence level, point anomalies represent single isolated events, contextual anomalies are only anomalous given their surrounding circumstances, and collective anomalies emerge from a group of instances that together form an irregular pattern[8]. Shifting to appearance, textural anomalies are characterized by surface-level deviations such as cracks, holes, and scratches, while functional anomalies impede function, such as missing or misaligned components.[2][3] Finally, at the extent level, local anomalies occupy only a small region of the image such as a lesion within healthy tissue, whereas global semantic anomalies span the image as a whole[1]. Together, these categories reflect that anomalies must be considered within context and at varying scales. More information about these types of anomalies are found under section 4.1.

7.2 Dataset

This section focuses on investigating datasets with their limitations and the impact the choices made for model architecture. Specifically the following will be discussed; the gap between real world scenarios against idealized datasets in terms of diversity, representation, and over-saturation. Thereafter is discussion on how the data shapes the model design choice.

7.2.1 Data Quantity and Quality Limitations

Among the papers reviewed, I noticed a plethora of different datasets being used for evaluation and model construction. Upon further investigation, the contributing factor to the abundance of datasets with overlapping domains is the result of limitations within datasets motivating researchers to develop new datasets that mitigate those shortcomings. Limitations often deal with two super categories, quality and quantity. Generally speaking, more high-quality data that represents the intended target distribution is better. For instance insufficient variations in object types and unrealistic anomalies do not represent the real world which consequently encourages authors to develop more challenging datasets [39]. The sentiment for increasing diversity and variation is also echoed by others [50]. Moreover, authors may seek to pioneer a new search space by providing a high quality, large scale dataset that was previously absent [43]. From a different perspective, authors have introduced new dataset for their own use case to validate to performance and behavior of the model under controlled settings [13].

With regards to data quality, Cui et al. has mentioned that existing open datasets cover a limited number of scenarios which are not comprehensive enough and do not capture the nature of actual industrial diversity. Furthermore, the simplicity of data in datasets does not accurately

reflect the complexities of industry [3]. This sentiment is mirror by Tao et al. who lists that some datasets are well imaged and uniformly illuminated which can inflate performance numbers as they do not mimic real complex industrial scenarios [2].

For popularity among the seminal works, it is clear that MVTecAD[43] has made a profound impact in the industrial manufacturing domain and may have been a contributing factor to the popularity of survey and research papers in that field. This conclusion can be applied for CUHK Avenue[47], UCSD Pedestrian[48], and STC[50] datasets. CIFAR (2009)[54], ImageNet (2009)[52], MNIST(1998)[46] being general datasets has seen a bit of usage when factoring in their date of publication. However, due to the popularity, MVTecAD has been over-saturated and methods such as PatchCore, FastFlow, AnomalyDINO, MaskVIT, DRAEM, Cutpaste, and more have achieved state-of-the-art performance at $\geq 98\%$ AUROC for both pixel (segmentation), and image (binary classification) levels. However, there is considerably lower performance on datasets such as CIFAR or VisA. As an example, the binary classification AUROC scores on these datasets are as low as 66.7% for fastflow[30] on CIFAR-10 and 79.5% for DRAEM[25] on VisA. This suggests that near-perfect scores on MVTecAD reflect saturation of that benchmark, and that evaluating across multiple benchmarks gives a more holistic view of model performance.

7.2.2 Data Influence on Models

Beyond general lack of data in terms of image count and video runtime [6][8], data limitations also deals with the nature of anomaly detection, which is the scarcity of anomalous data, especially for each researcher's domain of interest [3][11]. For industrial settings where "defects typically follow a long-tailed distribution"[4], this defect rarity issue is mitigated by leveraging the following techniques: pre-trained networks, reconstruction, and data generation [4]. pre-trained networks which are trained on a generalized datasets include models such as Wide-ResNet, ResNet, EfficientNet, etc. which theoretically results in the fundamental understanding of features like basic textures, shapes, and amalgamations forming distinct classes of everyday objects. This technique cuts down on the needs to train a model to learn from scratch thereby requiring less overall normal and abnormal data. However, it has been noted that this approach could negatively affect generalization abilities for different domains[3]. Reconstruction-based methods address the rarity problem by bypassing the need for anomaly data entirely. Rather than learning from anomalous examples, they learn the distribution of normal data, allowing anomalies to be identified by their failure to be reconstructed. Another approach is to augment anomalous data with fake data via data generation techniques. Anomalies generated through this method manifest as transformations to the image, cutting and pasting patches from and images and shuffling them around the image, or via natural synthetic anomalies.

7.3 Data and Model Assumptions

Every anomaly detection method relies on a set of assumptions about the underlying data distribution, and their performance can degrade sharply when those assumptions are violated. As an example, statistical-hypothesis-based methods for hyperspectral anomaly detection rest on the assumption that "background pixels follow a specific probability distribution, while anomalous pixels deviate from it."[5]

7.3.1 One Class

Perhaps the most pervasive assumption in the field is that anomaly detection is a one-class learning problem. Zaheer et al. articulated this assumption explicitly, noting that "anomaly detection

problems are based on learning only normal data distribution and anything occurring outside that distribution is termed as anomaly. It can be seen as a one class problem, in which all other classes are unknown."[6] That notion is also shared by Duong et al.[9]. This framing is largely a consequence of practical necessity. As Zaheer et al. explain, "all the possible anomalies in a given environment cannot be predicted. Consequently, a training data based on abnormal examples cannot be generated. Hence, majority of the anomaly detection systems are based on the idea of learning a normal distribution against which test examples are matched to find anomalous events."[6] This assumption introduces a fundamental epistemic limitation: the model can only define anomalies in terms of deviation from learned normality, without any explicit understanding of what constitutes a meaningful anomaly versus a benign variation. This becomes particularly problematic in situations where small variances are acceptable, or cases where situational context is required to determine anomalous events.

7.3.2 Reconstruction

For instance, the usage of autoencoders and GANs in reconstruction methods operates under the intuitive assumption that learning on normal data will fail to reconstruct anomalous outputs as anomalous data is never learned. However, this theory has come under contention as Cui et al. states "reconstructors like autoencoders and GANs are highly generalizable and robust, and the anomalous part of the image can be well reconstructed, leading to hypothesis failure." [3]. Furthermore, this corroborated by Tao et al. who demonstrates that an anomalous scratch mark defect can actually be reconstructed in a slightly different form due to the over-generalization of the autoencoder [2]. Techniques such as MemAE[18] and RIAD[24] support this claim where the generalization has caused the reconstruction of anomalies. This issue occurs when the assumption about the data is violated. The data must contain a high degree of correlation and structure to be bottlenecked into a lower dimensional space[2]. Ideally the bottleneck preserves important features by compressing correlated patterns, while noise and low-salience features are discarded. An abundance of dissociated, overly complex features that do not correlate well require a larger latent space to map. Excessively increasing the bottleneck will cause the model to overfit and generalize to more features, which is the bias-variance trade off. Another perspective is to favor learning the template of normality and avoid learning visual primitives such as edges, textures and color gradients, which could contribute to an anomalous reconstruction. Research has continued in this area as Li et al.[4] highlights the work of Zhang et al. who proposed a DC-AE anomaly detection framework, "designed to address the issue of traditional autoencoders mistakenly reconstructing anomalies as normal features[4]." Generative adversarial approaches assume that the generator can sample data from the manifold of training data and thereby only reconstruct normal data from the manifold of normal data[3]. In practice, the assumption is fragile under low-data regimes. Tao et al. observed that CycleGAN-based methods suffer because "the lack of data samples poses a challenge in training two GAN networks well at the same time in industrial scenarios[2]."

7.3.3 Pre-trained

Methods like PaDiM that utilized a pre-trained network operate under a different set of assumptions. PaDiM[28] fits a multivariate Gaussian to per-location features; this formulation assumes that the dataset is spatially aligned such that the same pixel coordinate corresponds to the same semantic content across training images [2]. Assumptions such as this limit the deployment applicability in scenarios where the same object is positioned or posed differently. Normalizing flow methods assume that feature representations can be mapped via an invertible transformation to

a known, tractable distribution, allowing exact likelihoods to be computed[2]. From this follows the anomaly detection assumption: normal samples get high likelihoods while anomalies get low ones. This works well when the flow is applied on top of a pre-trained network, since the resulting features form a cleaner distribution that the flow can actually fit[2].

7.3.4 Neighborhood

Santhosh et al.[8] notes that distance-based approaches assume normal data occupy dense neighborhoods. This assumption that normal points are dense and anomalous points sparse breaks down in two opposite cases. The first is a false positive: legitimately normal points that reside in sparse regions of the distribution have few close neighbors and are misclassified as anomalies. The second is a false negative: anomalies that co-occur or cluster together become locally dense and fall below the anomaly threshold. When reframed, this same vulnerability extends to any method that defines normality through structural conformity whether by neighborhood, cluster membership, or learned manifold.

7.3.5 Visual Distinctness

Another assumption mentioned is visual distinction. In relation to VAEs, the assumption is that, "anomalies in images are always visually distinct from the background or normal regions. This may not always be the case, as certain types of anomalies may be visually similar to the background, making them harder to detect using this method."[3] This is not always the case as low contrast, visually harmonious anomalies can camouflage defects. In hyperspectral imaging, an anomaly may be spectrally distinct but spatially indistinguishable; in industrial inspection, a hairline crack may register as little more than a few pixels of slightly altered intensity.

7.4 Drift

This section synthesis and discusses the various types of drift with regard to: normal behavior changing overtime, data used to train does not match data from real world, and annotation inaccuracies.

7.4.1 Concept Drift

Santhosh et al.[8] observed that "normal behavior evolves over time." This means that data fit to a historical distribution may become gradually miss-calibrated over time, this behavior is known as concept drift. Specifically defined as when the underlying mappings of input and output change typically over time, such that the same input may correspond to a different label than it did at training time. From another perspective, if an anomaly is defined as a rare occurrence, and that anomaly is frequently observed as occurring, then it no longer fits the definition of anomaly. The drift occurs when this happens over a period of time, otherwise the term shift may be more applicable.

The fundamental concept of continual learning addresses this issue directly, where the motivation is to enable life long learning capabilities while mitigating catastrophic forgetting of the prior distribution. Alternatively, the naive approach would be to retrain the network when the delta of drift becomes unacceptable. Taking PatchCoreCL[40] as an example, it maintains a per-task partitioned memory bank under a fixed global budget, re-applies coreset subsampling to preserve each task's feature coverage as allocations shrink, and uses a min-over-banks inference rule that performs implicit task identification at query time.

7.4.2 Covariate Drifts

Covariate or data drifts are finer-grained instances of domain drift. They are defined as the inputs to the network being different between training and test time while the model and real world distribution remains fixed. As an example, methods that assume a static scene or fixed object are highly susceptible to covariate drifts as slight changes in camera or object positions create new inputs that the model has never seen before. Another perspective, is that covariate shifts can serve as the anomaly signal for one class methods. If the model is only trained on normal samples, and a never seen before sample is introduced at inference time, it can induce a strong covariate shift (and drifts). The deviation can be used as the anomaly signal.

Methods that assume a stationary object or static scene may be subject to covariate (data) drift. This is caused by slight changes in camera angles, dirty lenses, or drastic scenery changes that do not match learned normal representation which will require recalibration via periodically retraining. Yu et al. frames these covariate drifts caused by fixed shutter speeds, light source intensity, image blur, etc. as interference factors[11]. Tao et al.[2] and Cui et al.[3] have partially shared similar sentiments.

An unfavorable instance of data drift is commonly addressed by multiple papers [5][3][8][11] and describes more of the situation of occurrence rather than a specific metric. The first issue is how benchmark and training conditions (typically normal conditions) do not match deployment, where datasets are too clinical and perfect which fails to represent actual production conditions. Models may exploit properties found in these benchmarks and training data thereby becoming overfit to the domain and failing to adapt to scenarios in real industrial environments[2]. This exploitation of certain features in the data can cause model performance variances from dataset to dataset. Second is the generalization argument: pre-trained networks are biased toward their training data and do not generalize well to different domains[3].

7.4.3 Annotator Drifts

Label drifts can be caused during dataset creation, as Tao et al. mentions that "Annotation noise may be inadvertently introduced when labeling the data."[2] This is further exacerbated by the lack of discussion on labeling criteria and agreement discussions for some datasets. [10]. However, some datasets such as UCF-Crime Dataset[39], DTD[51], Places 365[74] and LSUN[68] have addressed annotator drift mitigation strategies with annotator training and some require that annotators to achieve a 90% or higher accuracy on a controlled set of images to be accepted. Datasets like BMAD[77], which is a collection of externally sourced medical images, inadvertently address label drifts by relying on labels inherited from their original source, which is typically certified medical professionals. Furthermore, web-sourced and generated imagery based on search queries inherit their labels programmatically from the query or generation code.

7.5 Discussion of High Adoption of Certain Techniques

Through observation, it is evident in Table 2, and Figure 1 that representation, reconstruction, one-class classifications, and pyramid techniques are common approaches used in addressing vision related anomaly detection. The reasoning is straight forward where it applies to representation based techniques since the improvement in foundation models directly translates to an improvement for anomaly detection. The widely adopted use of one-class certainly favors this field as papers here tend to focus on the distinction of anomaly vs normality rather than the classification of a type of anomaly such as a tear, burn marks, etc. For pyramids, they tend to be popular due to the variety of abnormalities that can occur, from the pixel level to the contextual level (see Section 7.1

to learn more). Reconstruction based technique is another common method. It's popularity may be attributed to its intuitive approach and conceptual continuity from statistical method like PCA to Autoencoders which share similar principles. In practice, reconstruction can be split into two subcategories focused on rebuilding the entire image or the feature space representation of the image. This applies to both single images and sequential images for temporal data. It is notable that three papers in this category have used older methods like Structure Similarity Index Measure (SSIM)[20][24][15] directly. Moreover, a majority of papers still evaluate error based on L1, L2 similarities, or cosine distance. Among the reviewed papers in this category, the scoring aspect has established approaches, innovations tend to occur in the architecture itself. Therefore it may signify that the innovation tends to occur within the core architecture and not scoring metrics.

7.6 Usage of Techniques

This section provides guidance on when to use certain techniques that are found in Table 2, and Figure 1. **One-class** is the classification used to differentiate anomalous versus normal, while **multi-class** handles anomalous data with labels associated. **Memory banks** can be used to keep a collection of known normal samples for reconstruction methods or distance-based comparisons. The **reconstruction** technique is used when we want to generate a new anomaly free image given an anomalous image. **Prediction** techniques are used in conjunction with sequential images such that it predict future events based on prior images. Both **representation** techniques and **data augmentation** can be leveraged to address low training samples. **Pyramid** technique is used to handle anomalies at varying levels, such as pixel level versus semantic level. One can incorporate an **ensemble** technique to boost performance as each sub model is capable of capturing different features that the other models do not. The principle of **continual learning** allows for the introduction of new data such that the model adapts to data shifts. The **teacher student** technique can be used when we want to transfer the distribution from one model to another, typically from a larger model to a smaller model.

7.7 Comparative Discussion of Training Paradigms

Based on the surveyed papers, Table 2, and Figure 1 it is evident that a supervised method is not widely adopted within this field of study. This is because the majority of approaches follow the assumption that anomalies are rare and infrequent. Therefore it is not expected that anomalous data can be easily obtained which adjusts the balance of data to only favor images that depict normality. This is not say that there are no abnormal images in the dataset, quite the contrary. Abnormal images are used in evaluating the model's performance. This would suggest that supervised methods are a potential gap due to a count of one paper. However, contrary to that belief, models that would introduce labeled anomalous data tend to also generate their own anomaly signal via synthetic data. This is why there are a lot more papers classified under the self-supervised category. Unsupervised methods are the most straightforwards where all data is assumed to be normal and no annotations of the data are necessary. The thought is that if we capture the entire sample of normality it is easier than generating all forms of anomalies. This is to say that the space of anomalies is greater than the space of normality so it would be easier to just capture a set of normal examples. Zero-shot or training free loops seem to show promise where pre-trained networks already have highly generalized to the task. It is trivial to have a few normal samples for this generalized network to compare with test time samples. This zero-shot paradigm has potential to become apart of a pipeline that can continuously learn the distribution in an ever changing environment.

8 Future Directions

This section suggests creating datasets that reflect reality and aren't over saturated in usage. It further covers a potential exploration direction with data augmentation. Then it delves into performance gains by using pre-trained networks and how an improvement with a pre-trained network can have a positive impact on scores. Thereafter, the section highlights language model as another technique that can be a research direction. Lastly, under explored application domains discussed then potential in less popular anomaly detection scenarios.

8.1 Case Specific Datasets

Trends typically address the limitations of anomalies itself, particularly that anomalies are rare. Some methods like one-class classification, reconstruction, memory banks, etc. leverage this rarity when sampling from the distribution where a low probability density region equates to the flagging of anomaly. When viewed under a clustering perspective, anomalies are typically present in low-density neighborhoods which are distant from the well-formed clusters. In principle, this is exploiting covariate shifts to generate an anomaly signal. However, issues arise when the clusters and distribution are malformed due to misguided or overly simplistic training data[2][3]. As mentioned in section 7.2, the quantity and quality of a dataset is paramount to achieving a realistic evaluation of a model's performance. While shared datasets are useful for comparing various approaches, they tend to be idealized datasets, which do not mimic the variability of real world conditions, can over-inflate model scores without reflecting the domain drift that would appear in production. Therefore, it is recommended to use or develop a dataset that mimics the exact deployment scenarios with similar variability in features such as lighting conditions.

8.2 Data Augmentation

Data issues are further addressed via technical approaches such as data augmentations [11][3]. Approaches involve augmentation via rearranging patches of an image on a fixed grid, rotations, or synthetic textures generated from Perlin noise. The introduction of high quality realistic anomalous data by image generation models could induce a paradigm shift in image anomaly detection if proven to be highly effective. While some authors claim it is intractable to realistically cover all cases[6][9], improvements in synthetic image generation such as image inpainting models may be a future direction to bridge the gap. Furthermore, generating known anomalies for domain specific use cases [11] where the classification of distinct anomaly types can be easily done may prove useful.

8.3 Better Feature Representations

Foundations models are a cheap way to infuse machine learning method with fundamental knowledge, in a way it can be though of as common sense knowledge. This reduces the amount of training time required as any model that builds a network on-top does not need to be training starting from fundamental features like edges, shapes, or textures. As long as the feature vector output from a pre-trained network already has encoded features in a way that distinguishes the classes of interest then it can be trivial to have a classifier separate the data. As an example, if a car feature is present in the pre-trained network's latent space, then it is easy to build a car linear classifier. This idea is shown to work in practice by the authors of PANDA[33] who have stated that "combining pre-trained features with simple anomaly detection and segmentation methods convincingly outperforms much more complex, state-of-the-art methods". This idea is also corroborated by other methods like AnomalyDINO[37] which have highlighted strong visual features as a key

factor, SPADE[27] demonstrates state-of-the-art using a pre-trained network, and PaDiM[28] has shown strong performance using a pre-trained network and fitting a simple gaussian to the normal examples. This is opposed to constructing a overly engineered architecture. Moreover, based on the trend of using older Image Net based pre-trained networks to the DINO family of models with the continuation to Language Models; it can be expected that representation techniques are here to stay due to growing improvements. Therefore, better model internal representations is a potential region to focus on.

8.4 Foundation Models and Multimodal Approaches

Another method that provides additional signal without artificial data generation involves foundation models and the introduction of additional modalities. Foundation models, classified as representation-based in our taxonomy, are pre-trained on large general datasets and have proven tremendously popular, 15 papers in this review utilize the approach. Their broad generalization capabilities assist in downstream tasks and reduce the burden of collecting large amounts of domain-specific training data. Cui et al.[3], [4], and Cai et al.[1], each having surveyed the landscape, believe more research in this direction is warranted, including into pre-trained vision-language models. VLMs in particular offer a dual advantage: pre-trained weights that can be fine-tuned for specific tasks, and inherent multimodality that enriches feature representations. More broadly, the introduction of additional modalities beyond VLMs is also a promising avenue for improving detection performance [2] [42].

8.5 Under Explored Application Domains

Advancing discovery should not only be limited to model techniques, but domains as well. Based on visual anomaly detection survey papers, there seems to be a heavy emphasis on societal monitoring, inclusive of public safety, people and roads. Second most popular being manufacturing. The most popular type of dataset used by papers is MVTecAD (Industrial) and pedestrian walking (Social) datasets, which is to be expected as datasets were derived from seminal works which were derived from the survey papers. Techniques used in these fields have potential to be cross compatible with adjacent domains. Take for instance, social monitoring typically deals with a static background while foreground pedestrians or vehicles transverse across the scene. Abstractly this is akin to natural disaster monitoring and result in similar challenges such as lens degradation via dirt or lighting variations. Utilizing techniques from explored tasks and domains gives an opportunity to help develop non-predominant fields of study.

8.6 Use Case for Natural Disaster Monitoring

Returning to the motivation of this paper, natural disaster monitoring. The following justifies a potential use case for anomaly detection in this domain.

8.6.1 Framing as Anomaly Detection Problem

Anomaly detection can be a suitable candidate for natural disaster monitoring when placed under the correct framing. The variety of disasters are in abundance, disasters such as dense fog, heavy rain, ash plumes, wildfires, floods, earthquakes and more. To independently label and train a model on each disasters can be a time consuming process. However when framing it under the lens of anomaly detection, where normality is defined as the state without a natural disaster and

the natural disasters being anomalous, it is possible to transfer knowledge and techniques from anomaly detection field into the domain of natural disaster monitoring.

8.6.2 Contextual and Pixel Level Anomalies

Under that perspective, there are a few considerations to be had. First is how the noise signals are contextual. Lighting variations under cloudy weather and day night cycles are normal, but it becomes an anomaly if lighting variations are triggered by ash plumes that are out of the camera's view. Similarly, pixel level changes such as tree swaying or more semantic level changes such as animals appearing and disappearing can be considered normal. However when a crowd of animals emerges due to a wildfire, that would be anomalous.

8.6.3 Concept Drift and Temporal Aspect

Secondly, is that outdoor conditions evolve seasonally, thereby placing relevance on concept drifts and continual learning approaches. As an example of concept drifts, when a volcano initially erupts after a period of dormancy, that would be considered an anomaly. Now, after a considerable amount of time erupting, eruption becomes the norm. Would it be more appropriate to allow concept drift as a feature such that our definition of anomalous evolves over time such that the halting of eruption now becomes an anomaly or do we maintain a fixed view where we continue to believe that an eruption is anomalous regardless of any temporal aspect. Similarly, seasonal changes such as snowfall could be considered an anomaly or normal depending on how the task is defined. Would it be reasonable to consider the start of every season (the few days in the beginning) as an anomaly given its environment changes or should the model encompass a view where seasonal variations are normal? This becomes more complicated if there is a delay in seasonal changes as well. Perhaps general knowledge (from LLMs) is a prerequisite for such a system? Whether anomaly definitions should adapt over time or remain fixed is therefore task-dependent, and remains an open question for this domain.

8.7 Datasets and Data Augmentation

The absence of popular benchmarks for environment disaster imagery when it pertains to anomaly detection and combined with the rarity of events favors approaches that require few or no anomalous examples, such as representation, zero-shot vision-language techniques and data augmentation. If the task is restricted to a select few disaster types, synthetic data can be generated for each, sidestepping the need to cover the entire space of anomalies. This, however, trades away the ability to detect unanticipated events, so a hybrid that combines synthetic examples of known disasters with a one-class model of normality may offer the best of both.

Together, these observations position natural disaster monitoring as it pertains to anomaly detection as an under-explored domain where existing techniques provide a starting point but their underlying assumptions must be revisited.

9 Conclusion

Based on each author's perspective, pertinent to their domain and observations, there is no unanimous convergence of future directions. However, there are general patterns and consensus for which a subset of authors agree upon. The common applied approach is from the curation of the dataset with regard to quality, in practice, the dataset should have a similar distribution to its deployment

application while being cautious of the temporal aspect of normal data. Annotations applied to this dataset should be rigorously cross-checked. This task dependent dataset can be used to define normality. Next, involves building a pipeline. The main idea is that picking and choosing from the list of techniques presented in the taxonomy serves as a starting point for further innovation to occur. As a suggestion, various papers have outline the benefits of using pre-trained representation coupled with a pyramid technique for varying spatial resolutions and the addition of an anomaly classifier on top yields strong performance. However, it is prudent to be aware of some assumptions that certain techniques have made which have caused subpar performance. Alternatively, one may pursue innovations by incorporating the newer techniques, vanguarding experimentation with VLM and multi-modalities or by tweaking certain aspects of the pipeline such as developing a different loss function. Innovations can be further had by expanding on under-explored domains such as natural disaster monitoring. Anomaly detection provides a strong foundation when natural disaster detection is framed as a anomaly detection problem. Knowledge and techniques from this field can be directly transferable, however the limitations begin with how we define anomalous events in natural disasters.

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